# Integer-valued delivery promises: calibration, coverage and temporal shift

Oct 21, 2026 · @geniussuzane

## 1. Summary

If two forecasting models go through the same calibration and the same rounding to whole days, can their promised delivery dates be compared directly? This study finds that a matched procedure is not enough on its own, and shows where the comparison still breaks.

I started this project wanting to predict which Olist orders would arrive late. Along the way I realised that "late" compares two things: how long delivery takes, and how generous the promised date is. The question became: **when does a better forecast actually give a better promised date?** The data are public Olist e-commerce orders (Brazil, 2017–2018). The project's population keeps single-seller orders that were eventually delivered and whose Olist estimated delivery week falls on or before 23 July 2018; Section 3 explains why that last condition matters. A promise is a whole number of days; an order is covered if it arrives in time; the target coverage is 90%.

I compared a per-state quantile rule, a linear quantile regression and a shallow gradient-boosted model (HGB) under a protocol written down before any new model was fitted. The design matched the training data, the calibration target and the rounding rule. Three differences remained:

1. **The same calibration target did not give the same coverage.** Every calibrated policy aimed at 90% on the calibration period. On the evaluation period the linear policy covered 96.2% with 22.45 days and HGB 97.1% with 23.67 days, so their registered points compare different reliability levels. A post-hoc diagnostic that aligns coverage using the evaluation outcomes puts HGB 0.6–0.9 days below the linear model at the four coverage targets reported, in this sample and for this family of shift policies.
2. **The same rounding rule treats continuous and whole-day bounds differently.** Rounding the calibrated bound up adds less than a day of slack to a continuous forecast and none to the whole-day state rule: on the calibration period the continuous policies covered 91.1–91.2% against 90.3%. Rounding down instead keeps the coverage event of the bound (post hoc).
3. **The same evaluation sample is filtered on one comparator.** The cohort is defined by Olist's estimated delivery week, which removes most July purchases and all later ones. This limits what the comparison with Olist's own promise can show.

The registered verdicts stand: the matched design reversed the earlier regression-versus-rule judgement (+2.57 to −1.54 days), V selected no new linear setting, HGB did not clear the 1-day bar, and the registered policies did not meet the standard for replacing the existing rule.

The evaluation orders had already been examined earlier in the project, so this is a pre-registered exploratory study, not a blind test. Every analysis added after the scoring run is labelled post hoc and changes no registered verdict. This study examines when a matched evaluation procedure still produces comparisons at different reliability levels, and how calibration, integer rounding and cohort selection affect their interpretation.

## 2. Loss and decision: why a quantile, not the mean

If a late day and a spare day cost different amounts, the best promise is a quantile of the delivery time, not its mean. Let Y be the delivery time in whole days and d the promise, with the loss

$$\mathbf{L(d, Y) = c_late \cdot (Y - d)_+ + c_slack \cdot (d - Y)_+.}$$

Let R(d) = E[L(d, Y)] and F the distribution function of Y. Moving the promise from d to d + 1 saves c_late on every order with Y > d and adds c_slack on every order with Y ⩽ d. Therefore

$$\mathbf{R(d + 1) - R(d) = (c_late + c_slack) \cdot F(d) - c_late.}$$

This difference never decreases in d, so R is minimised at the smallest whole d with F(d) ⩾ q, where q = c_late / (c_late + c_slack). That is the empirical quantile rule the project uses (NumPy's inverted_cdf). The pinball loss at level q is this loss with c_late = q and c_slack = 1 − q (scikit-learn); a 90% target thus corresponds to a 9 : 1 penalty ratio. The mean minimises squared error and the median a 1 : 1 absolute loss; both treat a late day and a spare day alike.

**A check on validation data.** On the 12,045 validation orders I gave every order the same promise and computed the 9 : 1 loss for d = 1, …, 60. The minimum is at d = 20, the empirical 90% quantile: F(19) = 89.6% < 90% ⩽ F(20) = 91.4%. Promising the mean (11.43, rounded up to 12) would have covered only 65.5% of orders, at a loss of 23.76 per order against 15.61.

**What 9 : 1 does not mean.** It follows from choosing q = 0.90; I did not measure it, and it is not Olist's real cost ratio. Customer harm may not grow linearly with late days; if one late day already does most of the damage, a coverage constraint is closer to the truth than a linear cost. Therefore the study reports the trade-off between coverage and promised days. It does not claim a profit-optimal date.

## 3. Sample and comparison design

**Sample.** The population is defined on Olist's own estimated delivery week: an order enters only if that week lies between 27 February 2017 and 23 July 2018, its final outcome is delivered (on time or late), and it has a single seller. For the evaluation period E (purchases from 1 April 2018) this keeps 19,950 of 32,803 purchases. The week condition removes 5,175 of the 6,292 July purchases and every later one, so a late-window order enters only if Olist promised it a short time. The evaluation sample is therefore filtered on one of the comparators (Supplement S3c).

**Matched design.** Each calibrated policy used the same training orders, the same label cut-off, the same rule for seller-history features and the same calibration procedure. A matched procedure is not automatically neutral: rounding the calibrated bound up treats a continuous forecast differently from a whole-day rule (Section 5). The uncalibrated controls (B1 and the retained rule B0) are kept apart. A label counts only if the order was delivered before the decision date.

| Segment | Purchases | Orders | Late rate* | Role |
|---|---|---|---|---|
| F0 → V | Jan–Jun → Jul–Sep 2017 | 13,616 / 12,045 | 3.89% / 3.38% | fit and select settings |
| F | Jan–Sep 2017 | 25,666 | 3.67% | final fit |
| C | Oct–Dec 2017 | 17,056 | 8.74% | calibration shift |
| E | Apr–Jul 2018 | 19,950 | 4.26% | final scoring run |

*Against Olist's own estimated date.

**Calibration.** Each calibrated policy is shifted by its own number computed on C: the k-th smallest of its residuals $y - \hat{y}$, with $k = \lceil (n + 1) \cdot 0.9 \rceil$ and $n = 17{,}056$. The promise is then rounded up: $\max\{1, \lceil \hat{y} + \text{shift} \rceil\}$. This has the form of split conformal, but its guarantee needs C and E to be exchangeable, and C had a much higher late rate than E (Barber et al.). Section 5 shows how this rounding treats continuous and whole-day predictors differently.

**Three meanings of "coverage holds".** The study uses three different statements about coverage, and they support different claims.

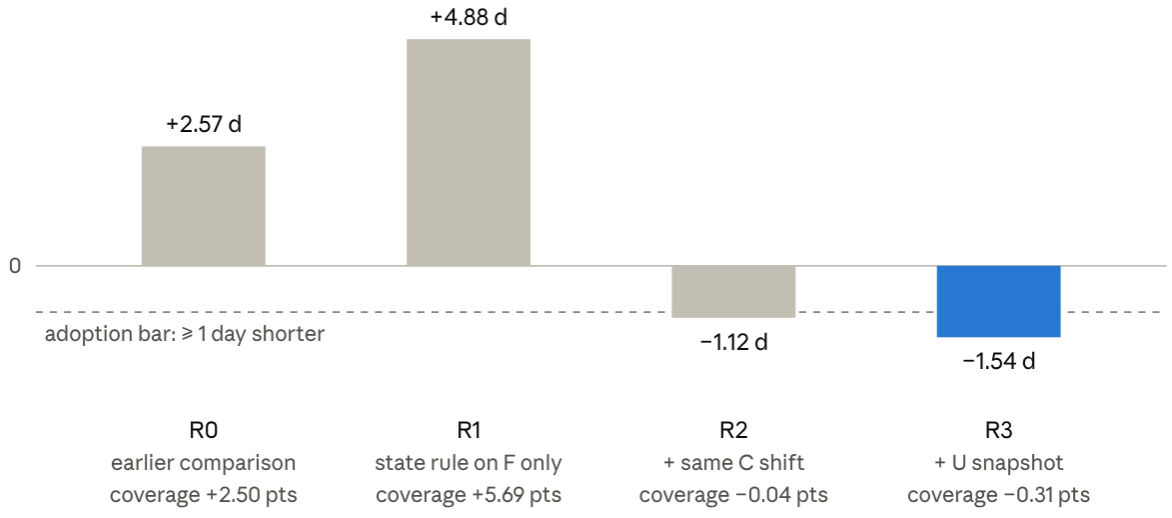
| What is observed or proved | What it shows | What it does not show |
|---|---|---|
| Coverage on C is at least k/n | A necessary consequence of the specified calibration rule | That the implementation is correct, or that future orders will reach the target |

| What is observed or proved | What it shows | What it does not show |
|---|---|---|
| Marginal coverage guarantee under exchangeability | $P(Y \leq U) \geq q$ on average over the random calibration set and the new order, if their scores are exchangeable | That every calibration draw, state or month reaches q. Exchangeability of C and E has not been established, so the guarantee is not asserted for E |
| Coverage of 96.2–97.1% on E | Empirical performance of the frozen policies on these selected historical orders | Reliability on all orders, or in future operations |

**Stepwise comparison.** The earlier comparison (R0) differed in three conditions at once, so I changed one condition per step. The pre-set primary comparison is R3 against R0; R1 and R2 are descriptive.

Matching the design flipped the gap from +2.57 to −1.54 days

Regression minus state rule, mean promised days on E (19,950 orders); one condition changed per step




outputs/e1/score/h1_steps.csv · E, April–July 2018, 19,950 orders

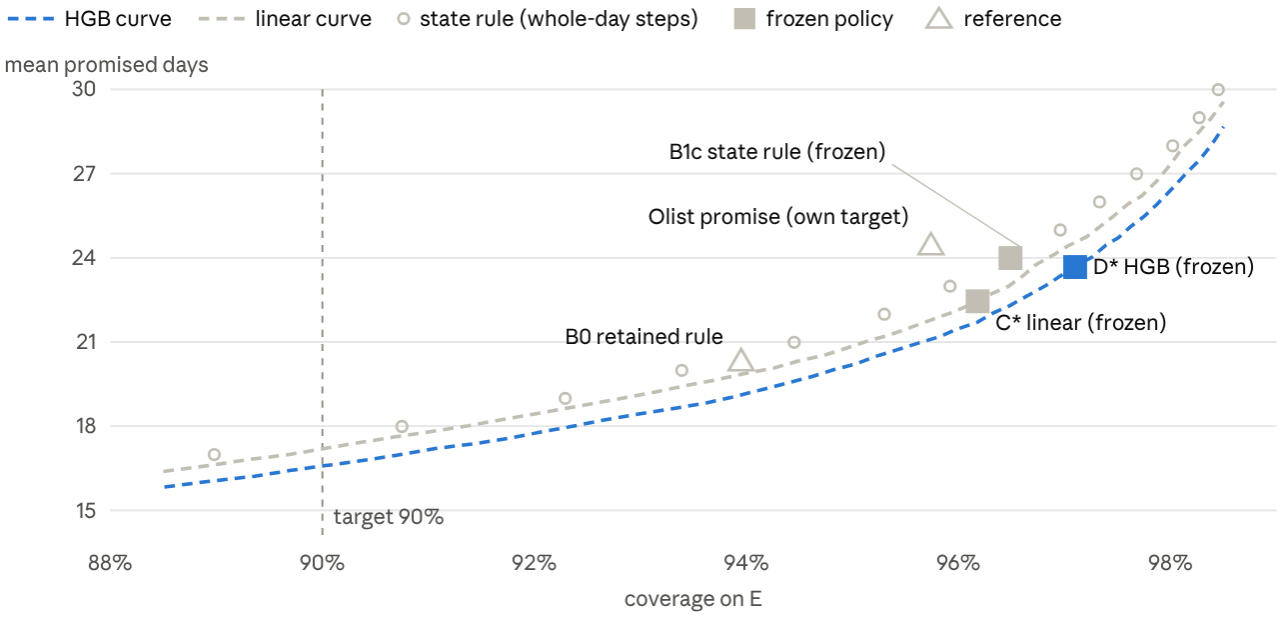
Under this fixed order, changing the design alone reversed the judgement; the largest step is giving the state rule the same calibration shift (+6 days). This is design sensitivity, not a causal statement about which method is better. Two references are kept apart: the matched reference B1c (registered under the name "fair reference": the state rule on F with the same calibration) and the replacement reference B0 (the project's retained rule, fitted on F + C without a shift).

# 4. Results

The E1 models and decision rules were frozen before the final scoring run on E, and each policy was scored once in that run. The figure keeps two kinds of evidence apart. Filled squares are those frozen policies, calibrated on C only. Dashed lines are diagnostic curves: each model's shift is moved using E's own outcomes, which aligns coverage after the fact. They describe the trade-off between days and coverage; they are not deployable policies and were not used to choose a model.

The frozen policies landed at different coverage on E

E, 19,950 orders. Filled squares: policies calibrated on C only, frozen, scored once on E.
Dashed lines: diagnostic curves whose shift uses E's own outcomes; descriptive only, not deployable, not used to choose a model.




outputs/e1/posthoc_m4/diagnostic_curves_E.csv and outputs/e1/score/policy_table.csv · E, 19,950 orders · dashed curves use E's outcomes, are post hoc and choose no policy

The frozen HGB policy covered 97.10% with 23.67 days, the frozen linear policy 96.18% with 22.45 days and B1c 96.49% with 23.99 days. Whole-day promises make coverage move in steps, so every diagnostic comparison reports the coverage actually reached rather than an interpolated match. At 90.02% HGB needs 16.60 days and the linear model 17.20; at 96.18% they need 21.71 and 22.44, while the state rule's nearest step is 96.49% at 23.99 days. In this sample and for this family of shift policies the HGB curve is the more favourable one; that is not a general ranking of the two models. On C at the registered shifts, HGB covered 91.14% with 24.54 days and the linear model 91.19% with 24.68.

The registered verdicts are unchanged. Intervals are week-block bootstrap intervals over 17 purchase weeks of unequal size with the models held fixed; they leave out refitting and selection uncertainty and are approximate. As a stability check (not a new interval), removing any one purchase week never changes the sign of a registered difference (Supplement S3e).

| Question | Comparison | Registered result | Verdict |
|---|---|---|---|
| H1: does the matched design change the judgement? | R3 vs R0 | gap moved by −4.11 days | met |
| H2: does a linear change help? | C* vs B3 | V selected no new linear setting, so no final comparison between different linear models was formed | recorded as "not supported" |
| H3: does shallow HGB help? | D* vs B1c | −0.32 days (about −0.55 to −0.17), coverage +0.61 pts | below the 1-day bar |
| Replace the retained rule? | C*, D* vs B0 | +2.15 and +3.37 days | standard not met |

At q = 0.9126, a target matching Olist's coverage on C, the linear policy promised 23.42 days at 96.62% coverage against Olist's 24.44 days at 95.74%. It passes the pre-set rule by 0.02 days, and a post-hoc interval for the day difference (−3.25 to +0.69) includes zero. Section 5 adds a further reason for caution: the evaluation sample is filtered on Olist's own estimated date.

## 5. Rounding, sample window and calibration period

Four features of the design move the registered points. All numbers below are post hoc: they reuse the frozen predictions, choose no new policy and change no registered verdict.

**Rounding to whole days.** The registered promise rounds the calibrated bound U = $\hat{y}$ + shift up. Because Y is a whole number, Y ⩽ U exactly when Y ⩽ ⌊U⌋, so rounding down keeps the coverage event of the untruncated bound, and the one-day minimum can only enlarge it. Rounding up adds slack only when U is not a whole number, i.e. for continuous forecasts. Rounding down would shorten each continuous promise by exactly 1 day on E and leave B1c unchanged. Coverage on C would fall from 91.1–91.2% to 90.01%, which still meets the rank bound, and HGB would be 1.32 days shorter than B1c at +0.18 points. A plain "subtract one day" is not coverage-preserving in general: it can exclude orders that sit exactly at a whole-number bound (Supplement S3b). The registered rule is therefore conservative rather than wrong, and the registered H3 verdict stands.

**Evaluation window.** July contributes 1,085 orders, with Olist promises of at most 26 days (56–63 in April–June) and a mean actual time of 6.2 days. In April–May alone, coverage is 95.1% for the linear policy, 96.4% for HGB and 95.5% for B1c; in June–July it is 98.3–98.5%.

Coverage is higher in the later, more heavily selected months. This is consistent with a contribution from the window restriction, but the analysis does not separate it from changes over time or in the order mix. The comparison with Olist is also filtered on the comparator itself.

**Calibration period.** C runs from October to December 2017. Late rates were 4.2%, 12.5% and 7.4% by month, and November holds 42% of C, so a shift learned there may over-cover a calmer period. This is a candidate explanation; a high November late rate does not by itself show that it caused the over-coverage.

**Replacement standard.** A policy may lose up to 1 point of coverage but gets no credit for extra coverage. At B0's own coverage (93.95%) on E, the diagnostic curves need 19.12 days for HGB and 19.83 for the linear model, against 20.31 for B0. The verdict says that the registered policies missed the standard, not that the models lack decision value.

These four points do not overturn the registered results. They change how the results should be read: each registered difference mixes forecast quality with calibration level, rounding and sample window, and this study does not separate those parts.

## 6. Limits and conclusion

**Limits.** One evaluation period, already examined earlier in the project and filtered on Olist's estimated week; a conditional population (single-seller, eventually delivered orders); a small stepwise model search; approximate intervals from 17 week clusters; no separate test of seasonality. Promised dates do not change logistics, so nothing here shows faster delivery or happier customers.

**Conclusion.** The pre-registered verdicts stand: the matched design reversed the regression-versus-rule judgement, V selected no new linear setting, HGB did not clear the 1-day bar, and the registered policies did not meet the replacement standard. The sensitivity checks narrow what these verdicts mean. The frozen policies reached different coverage, rounding up gave continuous forecasts extra slack, and the evaluation window was filtered on Olist's own dates. The next step is a small, separately recorded methods study: how to turn a continuous calibrated bound into a whole-day promise (rounding down and its boundary cases), and a paired simulation that separates rounding, temporal shift and selective labels. Online recalibration is left for later, because Olist labels arrive late and the first orders to be delivered are not a random subset.

A three-page mathematical note (why a quantile, and what k/n, k/(n+1) and the evaluation coverage each show) is attached separately. Supplementary material (model selection, the simulation, all post-hoc sensitivity tables), the integrity record and a contribution statement: `Supplement` (a separate PDF by email).